Contrastive Scene Change Representation Learning for High-Resolution Remote Sensing Scene Change Detection

Jue Wang, Yanfei Zhong, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Scene change detection (SCD) involves recognizing the high-level semantic change types for a bitemporal remote sensing image scene pair. Previous methods have implicitly considered joint bitemporal features as a change representation to perform SCD. However, they have not effectively enhanced the discriminative power of the change representation. Consequently, these methods struggle to recognize scene changes with both intraclass variation and interclass similarity. In this article, scene change contrastive (SCC) learning based on contrastive learning is proposed to ensure that bitemporal features are discriminative change representations for SCD recognition. Contrastive learning can learn specific discriminative features by gathering predefined specific positives and separating negatives in the projection space. In the SCC learning, the change representations, which are represented by the joint bitemporal features, are mapped to real and pseudo multiview change (MVC) projections by the proposed MVC projector and the pseudo change augmentation (PeCA) strategy. The change projections are then guided to be discriminative, exploiting both local spatial and global information. By doing so, the joint bitemporal scene features become more discriminative change representations, which enable accurate recognition of scene changes. The extensive experimental results obtained on two public datasets consistently demonstrate the effectiveness of the proposed method. The code is available at https://github.com/wdczs/SCC.**

***Index Terms*— Contrastive learning, multitemporal remote sensing images, scene change detection (SCD), scene classification.**

I. Introduction

CHANGE detection is an important technique to observe Earth's surface, to understand human activities and natural phenomena for decision making [1], [2], [3]. Scene change detection (SCD) is aimed at interpreting the high-level semantic changes at the scene level between bitemporal remote sensing images, which can serve for urban planning and land management applications [4], [5], [6]. As shown in Fig. 1(a), bitemporal images t_1 and t_2 are captured at the same geographical location by satellite at two different time points. Recognizing the change type of the bitemporal scene pair $\{t_1, t_2\}$ correctly means that bitemporal images t_1 and t_2 are classified correctly and simultaneously [5], [7]. As shown in Fig. 1(b), scene changes of bitemporal remote sensing images typically have intraclass variation and interclass similarity [4], [8], [9], [10], which makes it difficult to recognize scene changes accurately.

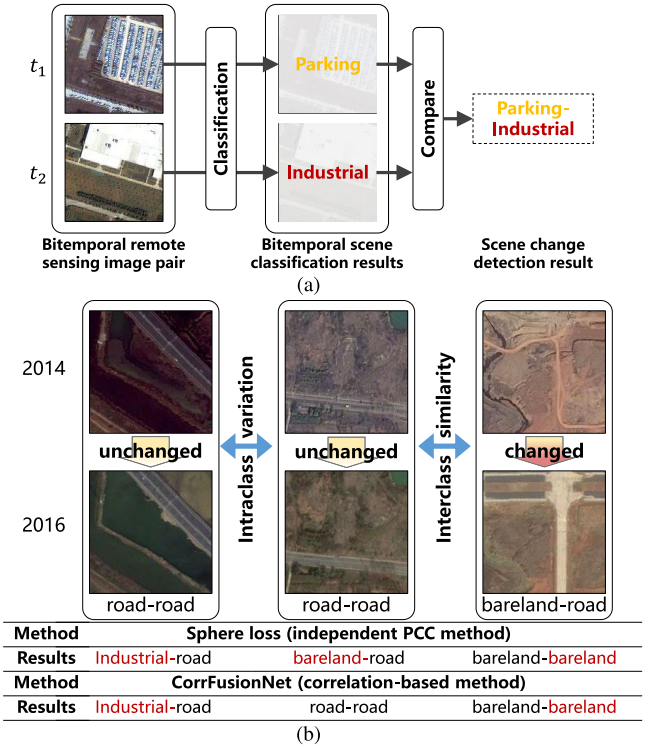


Fig. 1. (a) Example of SCD. (b) Illustration of the scene changes of intraclass variation and interclass similarity. The examples are from the WHMAVS dataset [7]. Sphere loss [11] and CorrFusionNet [5] are the advanced deep metric learning-based scene classification methods based on the independent PCC approach and temporal correlation-based SCD, respectively. The incorrect scene change predictions are highlighted in red.

Previously, researchers utilized hand-crafted features to extract information from bitemporal scenes for SCD [12], [13]. Because of the strong feature extraction ability of convolutional neural networks (CNNs), researchers have recently worked on using CNNs for SCD [5], the methods of which can be divided into two main categories: independent postclassification comparison (PCC) methods and temporal correlation-based methods. As shown in Fig. 2, the

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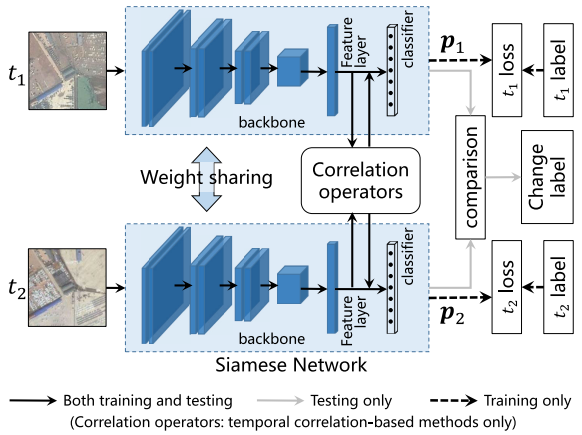


Fig. 2. Illustration of the existing SCD methods. $\boldsymbol{p}$ denotes the scene classification probability. The independent PCC method refers to the Siamese network without any communication between bitemporal classification, while the temporal correlation-based method refers to the Siamese network with the correlation operators.

independent PCC methods are based on an intuitive idea, in that they regard the SCD task as independent bitemporal scene classification [7], [12]. The naive independent PCC methods use a standard well-designed CNN (such as ResNet-50 [14]) with a feature layer to extract bitemporal feature vectors, and use a classifier to perform the bitemporal scene classification [7]. However, due to the complexity of bitemporal remote sensing scenes [4], [8], [9], the SCD performance of the naive independent PCC can be significantly limited by the feature discrimination [5]. There are two types of CNN-based scene classification methods that aim to learn discriminative features and can be applied to the independent PCC approach for SCD [9]. These include feature fusion-based methods [15], [16], [17], [18], [19], [20] and deep metric learning-based methods [11], [21], [22], [23]. The feature fusion-based methods fuse either multi networks' features [15], [19], [20] or global and local features of a single network to represent a scene image [16], [17], [18]. The deep metric learning-based methods aim to reduce intraclass distance and increase interclass distance in the scene feature space [11], [21], [22]. Even so, without modeling the temporal relationship between bitemporal images, all the independent PCC methods lack a mechanism to ensure that the bitemporal scene classification is correct simultaneously, leading to false alarms and false negatives (FNs) for the scene change. Recently, as shown in Fig. 2, based on canonical correlation analysis (CCA), temporal correlation-based methods [5], [24] have been proposed to model the temporal interrelationship between the bitemporal images within each pair to learn correlated components, which are used to improve the bitemporal feature discrimination before the features are input into the classifier [5]. Due to the large proportion of unchanged pairs, the correlated components can exhibit a strong ability to recognize categories of unchanged pairs. However, these methods do not consider the correlated components between different bitemporal scene pairs, which limits their effectiveness in compacting feature distributions and improving scene feature discrimination within each class. As shown in Fig. 1(b), while the first and second scene pairs both fall under the category of "road–road," they exhibit notable differences due to the presence of uninterested water and vegetation changes, respectively. Similarly, the second and third pairs are categorized as change and no change, but the images include a significant proportion of bare land, which results in the "bare land-road" and "road–road" categories being highly similar. These unique characteristics have led to inaccuracies in the SCD predictions of existing methods.

In short, the existing methods cannot handle the following two challenges simultaneously: 1) learning discriminative features for scene images with intraclass variation and interclass similarity, to guarantee accurate single-temporal scene classification; and 2) making the bitemporal scene classification be correct simultaneously, to guarantee the correct SCD. These two points are analyzed from the difficulties of scene classification, but why do they cause difficulties in identifying various scene changes? Since the performance of scene classification is largely determined by scene features, a more fundamental question is how scene features affect SCD. These questions prompted us to further analyze the relationship between bitemporal scene features and scene changes. The final output of SCD is achieved by combining the results from the bitemporal scene classification, which is derived from the weight-shared classifier utilizing bitemporal scene features. This means that the SCD result is actually determined by the combination of a bitemporal feature pair, which actually serves as a change representation. Therefore, the key challenge faced by the existing SCD methods is the insufficient discriminative power of the change representation, which is described by the joint bitemporal features and fails to differentiate between different types of scene changes.

Based on this finding, it is not difficult to imagine constructing a discriminative latent space for scene changes, where similar scene change representations are closer to each other, and dissimilar ones are farther apart. In this article, a novel scene change contrastive (SCC) framework with SCC learning based on contrastive learning is proposed to learn discriminative change representations. Contrastive learning can define specific positives and negatives for an anchor to contrast in the projection space, to learn specific discriminative features [25], [26], [27], [28]. In the SCC framework, the SCC learning asks the network to contrast the change projections, which are mapped from the joint bitemporal features, i.e., the change representations. In this sense, the joint bitemporal scene features are guided to be discriminative, to recognize the scene changes accurately.

Our contributions can be summarized as follows.

1) The SCC framework is proposed to learn discriminative change representations, to identify scene changes accurately. The SCC learning is a crucial element of the SCC framework, serving as a regularization term for the bitemporal scene classification. In the SCC learning, the joint bitemporal features are first mapped to the change projections by the proposed multiview change (MVC) projector and the pseudo change augmentation (PeCA) strategy. The change projections are then contrasted in the same projection space to enable them to be discriminative and to enhance the ability of the joint bitemporal features to describe changes effectively.

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2) To exploit both the local spatial and global information for the change representations, the MVC projector is proposed to map the joint bitemporal features to MVC projections. In the MVC projector, bitemporal feature maps are mapped to local-view change projections that focus on capturing the local spatial information of the bitemporal scenes. Meanwhile, the bitemporal feature vectors are mapped to global-view change projections that focus on capturing the global information of the scenes. Therefore, by contrasting the MVC projections in the same projection space, the proposed approach captures both local and global information in the change representations, enabling accurate modeling of scene changes.

3) To increase the positives and the negatives for SCC learning, based on geographical characteristics, the PeCA strategy is proposed to expand the quantities and classes of the change projections. In each mini-batch of the training, the strategy ensures that the input bitemporal scene images are crossed to combine bitemporal scene images at different geographical locations to construct pseudo-scene changes. This enriches both the quantity and type of the change projections, so as to enhance the SCC learning.

The experimental results obtained on two public SCD datasets of high-resolution remote sensing images consistently show that the proposed SCC framework can achieve promising SCD performance.

The rest of this article is organized as follows. Section II introduces the two techniques that are closely related to this work. Section III describes the proposed SCC framework and SCC learning. Section IV presents the proposed method's quantitative and visualization experimental results obtained on two public SCD datasets. Section V presents the ablation experiments conducted to verify the effect of the different parts of the proposed SCC framework. Finally, our conclusion and future work are given in Section VI.

II. Related Work

A. Contrastive Learning

Contrastive learning is a representation learning method with the core idea of "learn to compare" [29]. Recent contrastive learning methods have performed the core idea by gathering an anchor and the positives while pushing away the negatives in the projection space [25], [30], [31]. According to the different definitions of the negatives and the positives in the projection space, the features learned by contrastive learning can have different characteristics, such as the image-level projections for image classification [25], [30], [32], [33] and facial expression [34], the region-level projections for object detection [35], and the pixel-level projections for semantic segmentation [26], [31] and remote sensing land-cover classification [36].

In representative unsupervised contrastive learning methods, an image is augmented twice to obtain its multiple views as a positive pair, while the other images are used as negative samples, both of which are then contrasted by minimizing the InfoNCE loss [30], [32], [34], [36] or cosine loss [33]. When performed on huge unlabeled datasets, unsupervised contrastive learning can effectively pretrain an encoder for the specific downstream task. However, these methods suffer from pushing features of the same class away, which limits the effectiveness of unsupervised contrastive learning [25].

To address this issue, supervised contrastive learning employs multiple views of images and semantic labels to define positives and negatives. The supervised contrastive learning uses multiple positives and negatives for an anchor to contrast in the projection, which can be regarded as generalized triplet loss [37] and accounts for the effectiveness [25]. Following this, some methods use contrastive learning with labels for pretraining, but this requires additional time and computational resources before performing the downstream task [25], [27], [38]. Yang et al. [28] used contrastive learning with labels to enhance the feature discrimination for partial domain adaptation. More similar to this work, pixel-wise contrastive learning [26] has been proposed to regard contrastive learning as a regularization term of training a fully convolutional network (FCN) [39] for semantic segmentation, to improve the pixel-level feature discrimination in an end-to-end manner.

Here, the most significant disparity between existing contrastive learning methods and the proposed SCC learning is their distinctive approach to label utilization. Specifically, the SCC learning regards bitemporal scene images as a whole to model change projections while simultaneously mapping the scene label space onto the change label space.

B. Change Detection Based on Contrastive Learning

Some researchers proposed a contrastive learning-based method for binary change detection (BCD), which refers to the fundamental task of locating change regions at either the pixel or object level for bitemporal remote sensing images [40], [41], [42]. In recent years, the remote sensing community has considered the BCD task as binary pixel classification using FCN to identify whether each bitemporal pixel pair is changed or unchanged [42], [43], [44]. One significant challenge associated with these approaches is collecting paired labeled bitemporal images for training since FCNs are data-driven.

To overcome the challenge, some researchers proposed unsupervised CL-based methods [45], [46], [47], [48], which pretrain an encoder for BCD on unlabeled multitemporal remote sensing images. However, as all these methods work based on the strong assumption that bitemporal images at different locations cover different land objects and bitemporal images at the same location cover the same object, they suffer from noise during pretraining. Recently, Chen et al. [49] and Wang et al. [50] proposed SaDL and CDContrast methods based on supervised contrastive learning, respectively. These methods can use labels to pretrain the encoder for the BCD networks without noise. However, these methods have mainly improved the encoder's ability to capture land-cover information, neglecting the constraint between bitemporal features and change representations. More recently, Jiang et al. [51] have proposed a self-supervised global–local contrastive learning (GLCL) framework, which directly contrasts change representations. However, this approach fails to model the explicit relationships between land cover and their change, which have been shown in recent studies [49], [50] to be crucial for accurate BCD.

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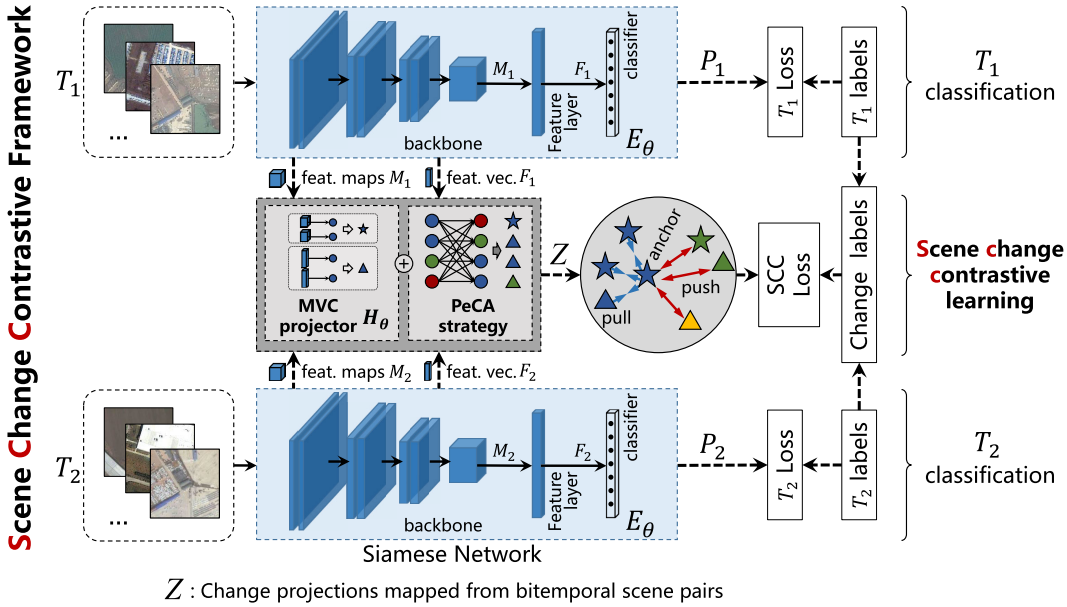


Fig. 3. Illustration of the proposed SCC framework and SCC learning. The global average pooling, which works between the backbone and the feature layer, is not shown. Bitemporal scene image sets T_1 and T_2 are first input into the Siamese network E_θ to extract bitemporal feature map sets, i.e., M_1 and M_2, and bitemporal feature vector sets, i.e., F_1 and F_2. These features are constructed to form bitemporal feature pairs, which are then mapped to real and pseudo-MVC projections by the proposed MVC projector H_θ and the proposed PeCA strategy. In this sense, the SCC learning is performed on the change projections, supervised by the change labels. It is important to note that both local-view and globe-view change projections can serve as anchors. The figure presented here illustrates an example of the SCC learning utilizing a local-view change projection as the anchor.

All the aforementioned methods are proposed for pixel-level BCD and do not take into account land-cover and change labels simultaneously.

III. Proposed SCC Framework

The SCC framework is proposed to make the joint bitemporal features discriminative change representations for the Siamese network to recognize scene changes. In the training phase of the SCC framework, the bitemporal scene images, their scene labels, and scene change labels are input into the Siamese network, which is then asked to perform bitemporal scene classification and SCC learning. In the testing phase of the SCC framework, the Siamese network assigns scene labels to the input bitemporal images. The bitemporal classification results are then compared and combined to obtain the scene change type of the input bitemporal scene pair.

A. SCC Framework With SCC Learning

Fig. 3 shows the pipeline of the proposed SCC framework with SCC learning. In the framework, a set of learnable parameters θ are used to define the Siamese network E_θ and the MVC projector H_θ. The Siamese network is made up of a well-designed backbone (such as ResNet [14]), followed by a global average pooling layer, a feature layer, and a scene classifier. The feature layer includes two fully-connected (FC) layers, to extract scene features. All these modules are weight-shared when the first and the second temporal images are input, respectively.

A mini-batch for the Siamese network includes N pairs of bitemporal scenes, i.e., the first temporal scene image set $T_1 = \{t_1^{(1)}, t_1^{(2)}, \ldots, t_1^{(l)}, \ldots, t_1^{(N)}\}$ and the second temporal scene image set $T_2 = \{t_2^{(1)}, t_2^{(2)}, \ldots, t_2^{(k)}, \ldots, t_2^{(N)}\}$, where $1 \leq l, n \leq N$. $t_1^{(l)}$ and $t_2^{(k)}$ denote the lth first temporal image and the kth second temporal image, respectively. When $l = k$, $\{t_1^{(l)}, t_2^{(k)}\}$ denotes the bitemporal scene images are at the same geographical location. When a mini-batch of N pairs of bitemporal scene pairs is input into the Siamese network E_θ, the τ-temporal feature map set M_τ output by the last convolutional layer of the backbone, the feature vector set F_τ output by the feature layer, and the τ-temporal scene classification probability vector set P_τ output by the scene classifier can be obtained as follows:

$$M_\tau, F_\tau, P_\tau = E_\theta(T_\tau) \tag{1}$$

where τ denotes the time phase, which can be 1 or 2. Here, $M_\tau = \{\boldsymbol{m}_\tau^{(1)}, \boldsymbol{m}_\tau^{(2)}, \ldots, \boldsymbol{m}_\tau^{(N)}\}$, where $\boldsymbol{m}_\tau^{(k)}$ denotes the τ-temporal k-th feature map. $F_\tau = \{\boldsymbol{f}_\tau^{(1)}, \boldsymbol{f}_\tau^{(2)}, \ldots, \boldsymbol{f}_\tau^{(N)}\}$, where $\boldsymbol{f}_\tau^{(k)}$ denotes the τ-temporal kth feature vector. $P_\tau = \{\boldsymbol{p}_\tau^{(1)}, \boldsymbol{p}_\tau^{(2)}, \ldots, \boldsymbol{p}_\tau^{(N)}\}$, where $\boldsymbol{p}_\tau^{(k)}$ denotes the classification probability vector of the τ-temporal kth image.

In the training phase, the SCC learning is performed in the projection space, the core idea of which is depicted in Fig. 4. To construct the shared projection space, bitemporal feature map sets M_1 and M_2 and feature vector sets F_1 and F_2 are mapped to N^2 local-view change projection set Z_{local} and the N^2 global-view change projection set Z_{global} by the proposed MVC projector H_θ and the proposed PeCA strategy, as follows:

$$Z_{\text{local}}, Z_{\text{global}} = H_\theta((M_1, M_2), (F_1, F_2)) \tag{2}$$

where $Z_{\text{local}} = \{\boldsymbol{z}_{\text{local}}^{(1)}, \boldsymbol{z}_{\text{local}}^{(2)}, \ldots, \boldsymbol{z}_{\text{local}}^{(N^2)}\}$ and $Z_{\text{global}} = \{\boldsymbol{z}_{\text{global}}^{(1)}, \boldsymbol{z}_{\text{global}}^{(2)}, \ldots, \boldsymbol{z}_{\text{global}}^{(N^2)}\}$. Here, since all the change projections are equally contrasted in the SCC learning, the N^2 local-view and N^2 global-view change projections can be combined into a change projection set Z, which includes $2N^2$

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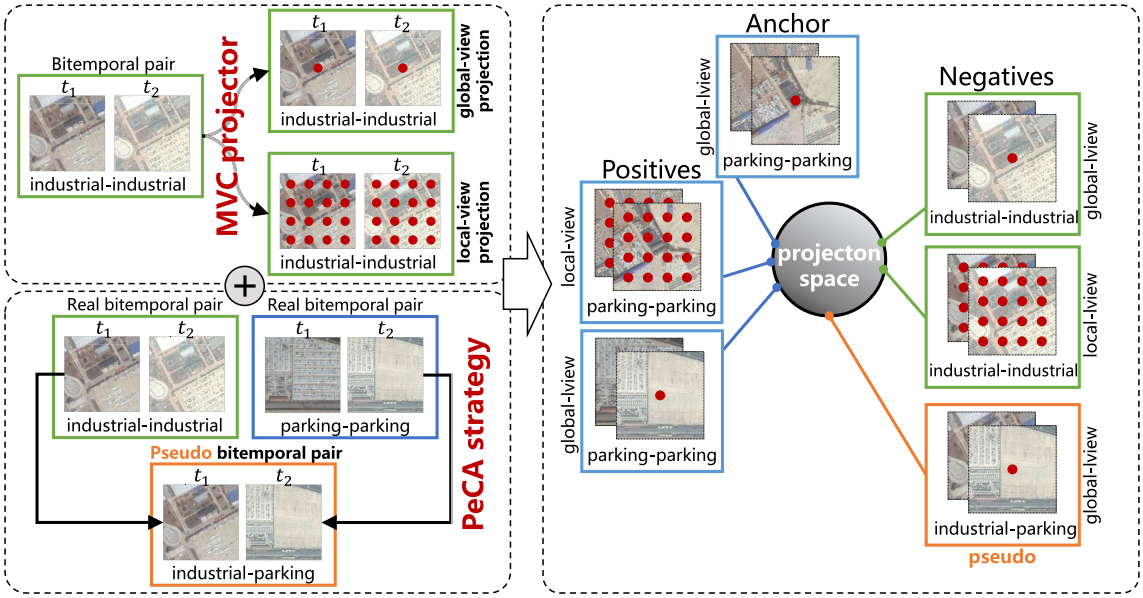


Fig. 4. Illustration of the proposed SCC learning, where the change projections are defined by local-view and global-view change projections, mapped from the joint bitemporal features of bitemporal scene pairs with change labels by the MVC projector. The global-view change projection focuses on globe information with globe average pooling (single red points), while the local-view change projection focuses on local spatial information (red-point matrices). The PeCA strategy based on geographical characteristics can increase the quantities and classes of the change projections.

change projections

$$Z = Z_{\text{local}} \cup Z_{\text{global}} \tag{3}$$

where $Z = \{z^{(1)}, z^{(2)}, \ldots, z^{(2N^2)}\}$. Here, $z^{(i)}$, which denotes the ith change projection in Z, is used to describe the change of bitemporal scene pair $\{t_1^{(l)}, t_2^{(k)}\}$ in the projection space. Within $2N^2$ of the local-view and global-view change projections obtained from a mini-batch of N bitemporal scene pairs, we let $I \equiv \{1, 2, \ldots, 2N^2\}$ be the index set of the change projections and $i, j \in I$.

In the SCC learning, both local-view and global-view change projections are given equal consideration. In this context, each change projection indexed by set I serves as an anchor and also functions as a positive or negative sample for the other projections, regardless of whether it is a local-view or global-view change projection. Based on contrastive learning theory [25], the SCC loss for the SCC learning is minimized to ensure that the change projections of the same change type are forced to be aggregated, and the change projections of the different change types are pushed away

$$\begin{aligned} L_{\text{SCC}} &= \frac{1}{2N^2} \sum_{i}^{I} L_{\text{SCC}}^{(i)} \\ &= \frac{1}{2N^2} \sum_{i}^{I} \frac{-1}{|\Phi(i)|} \sum_{\varphi}^{\Phi(i)} \log \frac{\exp\left(z^{(i)} \cdot z^{(\varphi)}/\upsilon\right)}{\sum_{j,j\neq i}^{I} \exp\left(z^{(i)} \cdot z^{(j)}/\upsilon\right)} \end{aligned} \tag{4}$$

where $\Phi(i) \equiv \{\varphi \in I(i) : y^{(\varphi)} = y^{(i)}, \varphi \neq i\}$ is the index set of all the positives in change projection set Z. Here, $z^{(i)} \cdot z^{(j)}$ represents the cosine similarity-based distance between two change projections, a measure that has gained widespread adoption in recent contrastive learning approaches [25], [30], [31], [32]. The primary advantage of cosine similarity lies in its bounded value range of $[-1, 1]$, which ensures greater stability during training when compared to the unbounded value range of the Euclidean distance measure ($[-\infty, +\infty]$) used in the classic contrastive loss [29]. $y^{(i)}$ denotes the scene change type of change projection $z^{(i)}$, which is formulated by comparing the corresponding bitemporal scene labels

$$y^{(i)} = \text{compare}\left(\boldsymbol{y}_1^{(l)}, \boldsymbol{y}_2^{(k)}\right) \tag{5}$$

where $\boldsymbol{y}_1^{(l)}$ and $\boldsymbol{y}_2^{(k)}$ denote the scene labels of the $t_1^{(l)}$ and $t_2^{(k)}$ images, respectively. $|\Phi(i)|$ represents the number of index elements of $\Phi(i)$, and υ represents a temperature parameter.

Essentially, minimizing the SCC loss is equal to asking $\exp(z^{(i)} \cdot z^{(\varphi)}) \geq \sum_{j,j\neq i}^{I} \exp(z^{(i)} \cdot z^{(j)}/\upsilon)$, which reduces the intraclass distance and increases the interclass distance of the change projections, to enhance the joint bitemporal features' ability to differentiate scene changes. Meanwhile, because the change projections are mapped from bitemporal feature map $\boldsymbol{m}$ and feature vector $\boldsymbol{f}$, the behavior of the change projections guides the positive scene features of the two temporal images to be aggregated, and the negative scene features of the two temporal images to be separated. This is further explained from the perspective of optimization in Section III-D.

A scene classifier is further trained to recognize the bitemporal scenes, so as to identify the scene changes in the test phase because the SCC learning does not include a classifier to predict the scene changes. In this context, the Siamese network is also constrained by the bitemporal scene classification loss L_{cls}

$$\begin{aligned} L_{\text{cls}} &= \frac{1}{2}(L_1 + L_2) = -\frac{1}{2N} \sum_{l}^{N} \left(L_1^{(l)} + L_2^{(l)}\right) \\ &= -\frac{1}{2N} \sum_{\tau}^{2} \sum_{l}^{N} \sum_{s}^{S} \boldsymbol{y}_{\tau}^{(l)}(s) \log \boldsymbol{p}_{\tau}^{(l)}(s) \end{aligned} \tag{6}$$

where L_1 and L_2 denote the cross-entropy classification losses on the T_1 and T_2 scene images. S denotes the number of

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scene classes. $\boldsymbol{y}_{\tau}^{(l)}$ denotes the label of the lth τ-temporal scene image.

In the SCC framework, to train the Siamese network, the joint loss L is formulated as

$$L = L_{\text{cls}} + \lambda L_{\text{SCC}} \tag{7}$$

where λ is the hyper-parameter to balance the classification loss and the SCC loss.

In the SCC framework's testing phase, the binary change result $\hat{y}_{\text{bi}}^{(*)}$ and semantic change result $\hat{y}^{(*)}$ of an arbitrary geographical location * is obtained by comparing the bitemporal scene classification results $\hat{\boldsymbol{y}}_1^{(*)}$ and $\hat{\boldsymbol{y}}_2^{(*)}$ of the bitemporal scene pair $\{t_1^{(*)}, t_2^{(*)}\}$, respectively,

$$\begin{aligned} \hat{y}_{\tau}^{(*)} &= \arg\max \boldsymbol{p}_{\tau}^{(*)} \\ \hat{y}^{(*)} &= \text{compare}\left(\hat{y}_1^{(*)}, \hat{y}_2^{(*)}\right) \end{aligned} \tag{8}$$

where $\boldsymbol{p}_{\tau}^{(*)}$ denotes the scene classification probability vector of the τth temporal image at geographical location *. Sections III-B and III-C will explain the proposed MVC projector and PeCA strategy.

B. MVC Projector in SCC Learning

To exploit both the local spatial and global information for the change representations, the MVC projector maps each bitemporal scene pair's feature vectors, and the last feature maps to the global-view change projections focusing on the global scope of the feature maps, and the local-view change projections focusing on the local regions of the feature maps, respectively. Traditionally, the feature vector of an image is obtained through the global average pooling of the feature map. While this captures the global information of an image, it comes at the expense of losing local spatial information within the feature map. Consequently, this approach makes it challenging to identify the discriminative regions necessary for distinguishing between scene categories and scene change categories [19]. Therefore, the local and global change projections are contrasted in SCC learning, to obtain precise representations of changes, which provides a comprehensive view of both local and global perspectives.

As shown in Fig. 5, the proposed MVC projector H_θ is made of a global change projector and a local change projector, which share the same learnable multilayer perceptron (MLP) h_θ. In particular, the local/global change projector uses the MLP and a concatenation operator to convert the bitemporal local/global representations to the local/global-view change projections.

To obtain the global-view change projection, bitemporal feature vector pair $\{\boldsymbol{f}_1^{(l)}, \boldsymbol{f}_2^{(k)}\}$ of input bitemporal scene pair $\{t_1^{(l)}, t_2^{(k)}\}$ are regarded as bitemporal global representations, because the feature vectors, which are the output of the top layer, are used to perform bitemporal scene classification by the classifier in the Siamese network. The global change projector is comprised of two weight-sharing MLP h_θ of "FC-ReLu-FC-ReLu-FC" architecture and a concatenation operator. In the global change projector, like the recent successful contrastive learning methods [30], [33], the bitemporal

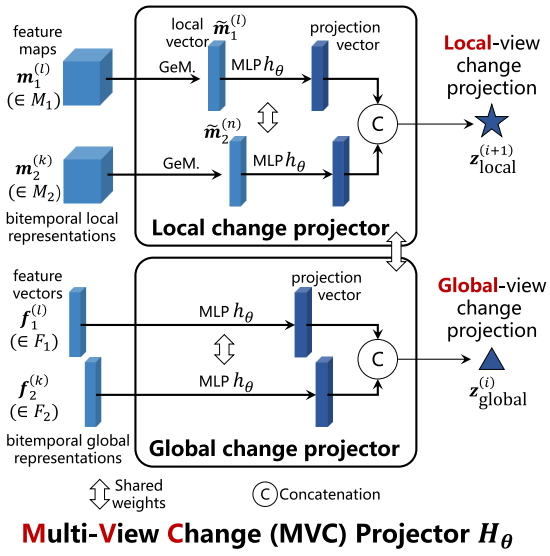


Fig. 5. Illustration of the proposed MVC projector. "GeM." denotes the GeM pooling. Bitemporal feature maps $\{\boldsymbol{m}_1^{(l)}, \boldsymbol{m}_2^{(k)}\}$ are mapped to the local-view change projection $\boldsymbol{z}_{\text{local}}^{(i+1)}$, while bitemporal feature vectors $\{\boldsymbol{f}_1^{(l)}, \boldsymbol{f}_2^{(k)}\}$ are mapped to the global-view change projection $\boldsymbol{z}_{\text{global}}^{(i)}$.

feature vectors $\boldsymbol{f}_1^{(l)}$ and $\boldsymbol{f}_2^{(k)}$ are first mapped to two 128-D projection vectors by MLP h_θ. The projection vectors are then concatenated to a 256-D global-view change projection $\boldsymbol{z}_{\text{global}}^{(i)}$

$$\boldsymbol{z}_{\text{global}}^{(i)} = \text{concat}\left(h_\theta\left(\boldsymbol{f}_1^{(l)}\right), h_\theta\left(\boldsymbol{f}_2^{(k)}\right)\right). \tag{9}$$

The bitemporal feature map pair $\{\boldsymbol{m}_1^{(l)}, \boldsymbol{m}_2^{(k)}\}$ of input bitemporal scene pair $\{t_1^{(l)}, t_2^{(k)}\}$ are regarded as bitemporal local representations to obtain the local-view change projection. To retain as much information as possible with the spatial dimension reduced, the feature maps are first converted to local vectors by generalized-mean (GeM) pooling [52], before being mapped to the local-view change projection vector in the local change projector. The GeM pooling is a generalized pooling between global average pooling and max pooling. The GeM pooling, which can extract feature vectors from the feature map, with the remaining local information that is useful for image retrieval but may not be useful for image classification, has been successfully used in the unsupervised image retrieval field. In this context, the GeM pooling is used to obtain the cth dimension of local vector $\tilde{\boldsymbol{m}}_\tau$ from the cth dimension channel of feature map $\boldsymbol{m}_\tau$ as

$$\tilde{\boldsymbol{m}}_\tau(c) = \frac{1}{H \times W}\left(\sum_{u}^{H \times W} (\boldsymbol{m}_\tau(c)(u))^q\right)^{\frac{1}{q}} \tag{10}$$

where q is the factor to balance the global and local information for the output local vector. When $q \to \infty$, the GeM pooling is max pooling, which focuses a lot on local information; when $q = 1$, the GeM pooling reduces to global average pooling. Clearly, the larger the value of q, the more attention the GeM pooling will pay to the local information. As shown in Fig. 5, bitemporal local vectors $\tilde{\boldsymbol{m}}_1^{(l)}$ and $\tilde{\boldsymbol{m}}_2^{(k)}$ are then mapped to the local-view change projection by the MLP h_θ and concatenation operation as

$$\boldsymbol{z}_{\text{local}}^{(i+1)} = \text{concat}\left(h_\theta\left(\tilde{\boldsymbol{m}}_1^{(l)}\right), h_\theta\left(\tilde{\boldsymbol{m}}_2^{(k)}\right)\right). \tag{11}$$

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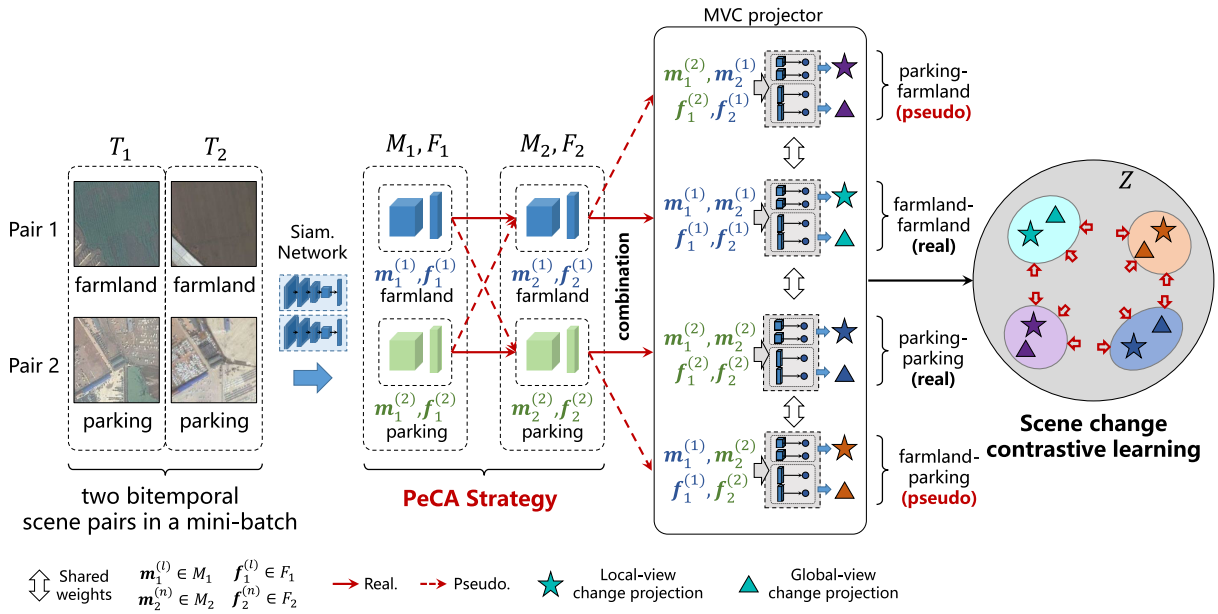


Fig. 6. Illustration of the proposed PeCA strategy. Different colors of change projections denote different scene change types. A mini-batch includes $N = 2$ pairs of bitemporal scene images that are mapped to $N = 2$ bitemporal feature vector sets F_1 and F_2 and $N = 2$ bitemporal feature map sets M_1 and M_2. Here, $F_\tau = \{\boldsymbol{f}_\tau^{(1)}, \boldsymbol{f}_\tau^{(2)}\}$ and $M_\tau = \{\boldsymbol{m}_\tau^{(1)}, \boldsymbol{m}_\tau^{(2)}\}$. The features are then crossed to obtain $N^2 = 4$ bitemporal feature vector pairs (i.e., $\{\boldsymbol{f}_1^{(2)}, \boldsymbol{f}_2^{(1)}\}, \{\boldsymbol{f}_1^{(1)}, \boldsymbol{f}_2^{(1)}\}, \{\boldsymbol{f}_1^{(2)}, \boldsymbol{f}_2^{(2)}\}$, and $\{\boldsymbol{f}_1^{(1)}, \boldsymbol{f}_2^{(2)}\}$) and $N^2 = 4$ bitemporal feature map pairs (i.e., $\{\boldsymbol{m}_1^{(2)}, \boldsymbol{m}_2^{(1)}\}, \{\boldsymbol{m}_1^{(1)}, \boldsymbol{m}_2^{(1)}\}, \{\boldsymbol{m}_1^{(2)}, \boldsymbol{m}_2^{(2)}\}$, and $\{\boldsymbol{m}_1^{(1)}, \boldsymbol{m}_2^{(2)}\}$), respectively. These feature pairs are then mapped to $N^2 = 4$ local-view change projections and $N^2 = 4$ global-view change projections, i.e., totaling $2N^2 = 8$ change projections by the proposed MVC projector. "Farmland-farmland" and "parking-parking" are change types of the real bitemporal scene pairs, which naturally exist in the mini-batch. In contrast, "parking-farmland" and "farmland-parking" are change types of the pseudo bitemporal scene pairs obtained by the PeCA strategy.

Because the local-view change projection $\boldsymbol{z}_{\text{local}}^{(i+1)}$ and global-view change projection $\boldsymbol{z}_{\text{global}}^{(i)}$ are both extracted from the same bitemporal scene pair of $t_1^{(l)}$ and $t_2^{(k)}$, the two kinds of change projections share the same change label, which can be obtained by comparing the bitemporal scene labels of $t_1^{(l)}$ and $t_2^{(k)}$, as in (5).

C. PeCA Strategy in SCC Learning

To increase the number of positives and negatives in the projection space to enhance the SCC learning, the PeCA strategy crosses the geographical locations of the feature maps and feature vectors of the input bitemporal images in a mini-batch, to increase both the quantity and categories of the change projections.

The core idea of the PeCA strategy is that bitemporal scene images at different geographical locations are combined to be regarded as pseudo-bitemporal scene pairs, while the bitemporal scene images at the same geographical location are regarded as real bitemporal scene pairs. Namely, bitemporal scene images $t_1^{(l)}$ and $t_2^{(k)}$ are defined as pseudo when $l \neq n$ and are defined as real when $l = n$, the change labels of which are all formulated by comparing the labels of the corresponding $t_1^{(l)}$ and $t_2^{(k)}$ as in (5).

In practice, the PeCA strategy can be performed on features rather than on images because the change projections are mapped from features extracted by the weight-shared Siamese network, so as to guarantee computational efficiency. As shown in Fig. 6, when bitemporal feature map sets $\{M_1, M_2\}$ and feature vector sets $\{F_1, F_2\}$ are first extracted from the N pairs of bitemporal images in a mini-batch, the PeCA strategy is then used to obtain all possible N^2 combinations of bitemporal feature maps and N^2 combinations of bitemporal feature vectors, which are mapped to change projections by the MVC projector. Moreover, because the change labels are obtained by the bitemporal scene labels, as in (5), the PeCA strategy can obtain new change types, compared to the mini-batch's existing ones.

In summary, the PeCA strategy expands both the change amounts and the change categories of the existing SCD datasets, so as to enhance the SCC learning.

D. Optimization

This section describes how we derive the gradients of the SCC learning, which reveals how the change projections influence the joint bitemporal features for describing changes and the single-temporal features for scene classification.

When the ith change projection $\boldsymbol{z}^{(i)}$ is a global-view change projection, its gradient involves both feature map $\boldsymbol{m}_\tau^{(l)}$ and feature vector $\boldsymbol{f}_\tau^{(l)}$ of the τ-temporal lth scene image, which can be formulated as

$$\frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{m}_\tau^{(l)}} = \frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{z}^{(i)}} \frac{\partial \boldsymbol{z}^{(i)}}{\partial \boldsymbol{f}_\tau^{(l)}} \frac{\partial \boldsymbol{f}_\tau^{(l)}}{\partial \boldsymbol{m}_\tau^{(l)}}$$
$$\frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{f}_\tau^{(l)}} = \frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{z}^{(i)}} \frac{\partial \boldsymbol{z}^{(i)}}{\partial \boldsymbol{f}_\tau^{(l)}}. \tag{12}$$

When the ith change projection $\boldsymbol{z}^{(i)}$ is a local-view change projection, its gradients only affect the feature map

$$\frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{m}_\tau^{(l)}} = \frac{L_{\text{SCC}}^{(i)}}{\partial \boldsymbol{z}^{(i)}} \frac{\partial \boldsymbol{z}^{(i)}}{\partial \boldsymbol{m}_\tau^{(l)}}. \tag{13}$$

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Because the local-view and the global-view change projection are equally considered in the SCC loss, as in (4), the gradient with respect to any change projection $z^{(i)}$ can be computed as follows:

$$\frac{\partial L_{\mathrm{SCC}}^{(i)}}{\partial z^{(i)}} = \frac{1}{\upsilon}\left\{\sum_{\varphi}^{\Phi(i)} z^{(\varphi)}\left(W^{(i\varphi)} - \frac{1}{|\Phi(i)|}\right) + \sum_{\mu}^{M(i)} z^{(\mu)} W^{(i\mu)}\right\} \tag{14}$$

where $M(i) \equiv \{\mu \in I(i) : y^{(\mu)} \neq y^{(i)}, \mu \neq i\}$ is the index set of all the negatives in the change projection set Z of a mini-batch, and

$$W^{(ix)} \equiv \exp\left(z^{(i)} \cdot z^{(x)}/\upsilon\right) / \sum_{j,j\neq i}^{I} \exp\left(z^{(i)} \cdot z^{(j)}/\upsilon\right) \tag{15}$$

where the term $0 \leq W^{(ix)} \leq 1$ and x can be the positive index φ or the negative index μ.

The combination of (14) and (15) shows that the movement of $z^{(i)}$ in the projection space is determined by the sum of the positive change projections, weighted by $W^{(i\varphi)}$, and the sum of the change projections of different classes, weighted by $W^{(i\mu)}$. This behavior gathers the same-type change projections and separates the different-type change projections. In this context, because the $L_{\mathrm{SCC}}^{(i)}/\partial z^{(i)}$ of (14) treats the bitemporal features as a single entity in (12) and (13), the bitemporal features are simultaneously guided by the same movements of the change projections, the ability of the joint bitemporal features to describe changes is enhanced. In addition, the close relationship between change projections and scene features enhances the discrimination of scene features, ultimately improving classification performance.

IV. Experiments

Two public SCD datasets—the Hanyang dataset [12] and the WHMAVS dataset [7]—were used to validate the proposed method. All the experiments were implemented in PyTorch [53] on an NVIDIA Tesla P100 graphics processing unit (GPU).

A. Experimental Settings

1) Benchmark Methods and Implementation Details: Following the research on SCD [5], [7] and scene classification [11], [21], three powerful networks, i.e., VGGNet-16 [54], ResNet-50 [14], and DenseNet-121 [55], were used as backbones in the experiments. Following [5], each backbone is accompanied by a feature layer that consists of two FC layers, each containing 1024 units and utilizing rectified linear unit (ReLU) activation functions, for the purpose of extracting scene features. In this sense, the baseline method is defined by a backbone with a feature layer and a linear classifier to perform the independent PCC method. The proposed framework was compared with the state-of-the-art SCD method of CorrFusionNet [5] as well as three advanced scene classification methods that aim to learn discriminative features: AGOS, which uses global and local feature fusion [17]; D-CNN [21]; and sphere loss [11], both of which employ deep metric learning techniques. All the methods recognize the scene changes by comparing the bitemporal classification results in the inference stage.

To be consistent with [5], all the Siamese networks were optimized by a mini-batch stochastic gradient descent (SGD) optimizer with the initial learning rate set to 0.001 and the momentum set to 0.9. The number of epochs and batch size were set to 100 and 32, respectively, while the weight decay was set to 1e-4. The FC layer's feature dimensions were set to 1024 for all the Siamese networks. The temperature υ was set to 0.07 as in [25].

2) Evaluation Metrics: The models were primarily evaluated using overall accuracy (OA) and kappa coefficient (KC) with respect to the number of classes, X. These can be denoted as

$$\mathrm{OA}(X) = \frac{\sum_{x=1}^{X} u_{xx}}{\sum_{x=1}^{X}\sum_{\hat{x}=1}^{X} u_{x\hat{x}}} \tag{16}$$

$$\mathrm{KC}(X) = (\mathrm{OA}(X) - \eta)/(1-\eta)$$

$$\eta = \frac{\sum_{x=1}^{X}\left[\left(\sum_{\hat{x}=1}^{X} u_{x\hat{x}}\right)\cdot\left(\sum_{\hat{x}=1}^{X} u_{\hat{x}x}\right)\right]}{\sum_{x=1}^{X}\sum_{\hat{x}=1}^{X} u_{x\hat{x}}} \tag{17}$$

where $u_{x\hat{x}}$ denote the number of samples that belong to the xth class and are classified as the $\hat{x}$th class. When S represents the number of scene classes, the number of scene change classes is S^2. In this context, the semantic SCD OA_tr and the semantic SCD KC were computed by $\mathrm{OA}(S^2)$ and $\mathrm{KC}(S^2)$, respectively, to evaluate the network's ability to recognize the land-cover/land-use change types of the bitemporal scene pairs. That is, only when the bitemporal scene classification results are correct simultaneously, can the scene change type of a bitemporal scene pair be considered correct. To evaluate the scene classification performance for the first time phase and the second time phase, $\mathrm{OA}(S)$ was used to compute OA_t1 and OA_t2, respectively.

Meanwhile, to evaluate the binary SCD, which is concerned with whether the bitemporal scene pairs can be correctly identified as change or not, OA_bi was computed by OA (2) with true positive (TP), FN, false positive (FP), and true negative (TN) involving changed and unchanged scenes.

B. Experiment 1: The Hanyang Dataset

The Hanyang dataset [12] includes bitemporal remote sensing images captured by the IKONOS satellite in 2002 and 2009 and covers the Hanyang region of the city of Wuhan, Hubei province, China. The bitemporal remote sensing images, which are of size 7200×6000 and of spatial resolution 1 m, were cropped to patches of size 150×150, so as to obtain 190 bitemporal image pairs for training and 1050 bitemporal image pairs for testing. For this dataset, 100 epochs mean 600 iterations. All these cropped images are annotated by eight land-use scene classes, which are 1—parking, 2—water, 3—sparse houses, 4—dense houses, 5—residential region, 6—idle region, 7—vegetation region, and 8—industrial region. This means that there are potentially 64 scene change categories. The distribution of the number of bitemporal scene categories is presented in Fig. 7. As can be observed from Fig. 7, the

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TABLE I

QUANTITATIVE RESULTS OBTAINED ON THE HANYANG DATASET. THE BEST AND SECOND-BEST ACCURACIES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Backbone	Method	OA_t1	OA_t2	OA_bi	OA_tr	KC
VGGNet-16	Baseline	81.33	84.57	84.76	71.71	68.81
	CorrFusionNet*	**86.29**	86.38	85.33	75.71	72.58
	CorrFusionNet	83.14	87.90	85.81	75.62	72.98
	AGOS	84.95	**88.76**	**87.81**	**77.14**	**74.69**
	D-CNN	84.10	86.10	85.43	74.38	71.71
	Sphere loss	81.71	86.29	85.71	73.33	70.41
	SCC (proposed)	84.95	87.52	86.76	76.38	73.88
ResNet-50	Baseline	80.76	87.05	84.10	72.19	69.11
	CorrFusionNet*	81.86	89.05	85.30	74.76	72.52
	CorrFusionNet	83.24	86.38	85.05	73.52	70.67
	AGOS	85.14	**90.00**	86.95	77.24	74.75
	D-CNN	86.10	89.43	86.67	78.00	75.60
	Sphere loss	81.14	87.52	84.86	73.24	70.25
	SCC (proposed)	**86.67**	89.24	**87.90**	**78.86**	**76.55**
DenseNet-121	Baseline	86.76	86.86	86.00	76.57	74.07
	CorrFusionNet*	86.42	**89.70**	88.13	78.85	76.73
	CorrFusionNet	87.33	88.86	88.19	78.95	76.67
	AGOS	86.57	86.38	86.48	76.10	73.50
	D-CNN	86.86	85.62	86.29	76.29	73.85
	Sphere loss	87.62	87.81	87.43	78.29	75.97
	SCC (proposed)	**88.67**	89.52	**88.48**	**80.48**	**78.36**

The CorrFusionNet method with "*" in the table indicates that the presented experimental results are cited from the original paper. "SCC" means the proposed SCC framework.

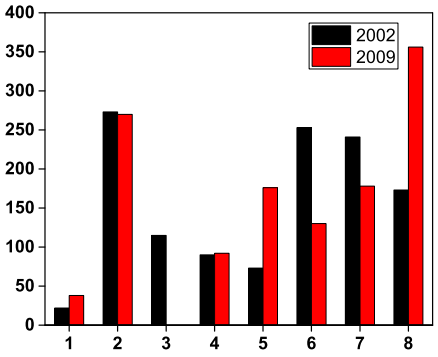


Fig. 7. Distribution of the number of bitemporal scene categories in the Hanyang dataset.

imbalanced scene distribution and limited data volume render the Hanyang dataset a challenge.

As shown in Table I, in the experimental results for each backbone, it can be seen that the proposed method can achieve promising scene classification accuracies and SCD performances. In particular, the proposed SCC framework with the DenseNet-121 backbone outperforms the other two backbones. In this sense, the proposed SCC framework obtains improvements of 0.29%–2.48% in OA_bi, 0.67%–4.19% in OA_tr, and 0.9%–4.51% in KC when compared with the other methods. It can also be found that the performance of AGOS with DenseNet-121 is inferior to its baseline. This outcome can be attributed to the introduction of the AGOS module, which added 6.23 million parameters that had to be trained from scratch using only 150 pairs of bitemporal scene

TABLE II

BINARY SCD RESULTS OBTAINED USING DENSENET-121 OBTAINED ON THE HANYANG DATASET. THE BEST AND SECOND-BEST ACCURACIES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

	TP	FN	FP	TN
Baseline	378	35	112	525
CorrFusionNet	384	29	95	542
AGOS	373	40	102	535
D-CNN	373	40	104	533
Sphere loss	**386**	**27**	105	532
SCC (proposed)	**386**	**27**	**94**	**543**

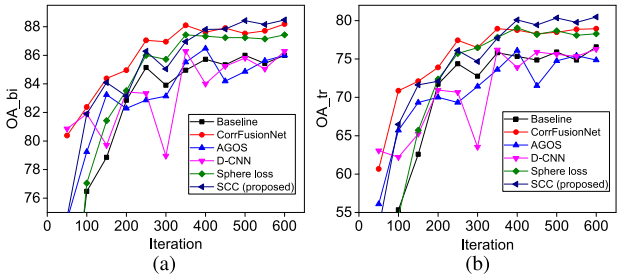


Fig. 8. (a) OA_bi and (b) OA_tr curves of the different methods on the test set of the Hanyang dataset, using the DenseNet-121 backbone.

images in the training set. Regarding the metric learning-based scene classification methods, it is interesting that D-CNN with DenseNet-121 backbone performs even worse than the baseline DenseNet-121 in Table I. This may be because D-CNN handles the feature distance only within each mini-batch, which tends to be unstable. In contrast, sphere loss learns the center for each scene class for all the bitemporal scene images, where the center can be regarded as a moving average on the whole dataset. As a result, sphere loss achieves great SCD performance in Table I.

When binary SCD is considered, as shown in Table II, the Siamese network trained by the proposed SCC framework achieves the best TP and TN scores, indicating its powerful ability to recognize changed and unchanged bitemporal scenes. Fig. 8 shows the test curve of OA_bi and OA_tr during the training, using the DenseNet-121 backbone. It can be found that the convergence of the proposed SCC framework is also fast and stable. Although the D-CNN method converges faster than the proposed SCC framework using the small-scale Hanyang dataset, the final SCD performance of D-CNN is worse than that of CorrFusionNet, sphere loss, and the proposed SCC framework.

The confusion matrices of scene classification results obtained by the two SCD methods, i.e., CorrFusionNet and the proposed SCC framework with DenseNet-121 backbone, are presented in Fig. 9. For CorrFusionNet, its errors are concentrated in the "parking" and "residential_region" classes, 20.00% and 9.84% of which are incorrectly classified as "industrial_region," respectively. Meanwhile, for the proposed SCC framework, no scene category of its classification results has an error rate of more than 20%, which means that the scene error rate of the proposed SCC framework is more balanced. This explains why the proposed SCC framework

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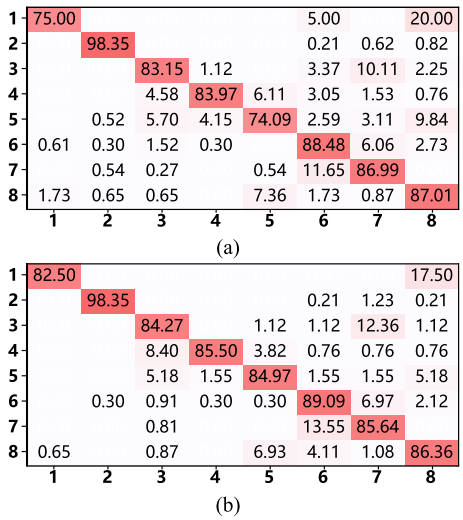


Fig. 9. Confusion matrices of bitemporal scene classification results on the testing set of the Hanyang dataset using (a) CorrFusionNet [5] and (b) proposed SCC framework, with backbone model DenseNet-121. 1—Parking, 2—water, 3—sparse_houses, 4—dense_houses, 5—residential_region, 6—idle_region, 7—vegetation_region, and 8—industrial_region.

obtains comparable OA_t1, OA_t1, and OA_bi scores to CorrFusionNet, but can outperform the CorrFusionNet by 1.71% in OA_tr and 1.84% in KC, respectively. The test set predicted by the proposed SCC framework with DenseNet-121 backbone is visualized in Fig. 10. From this, it can be observed that the majority of scenes and their changes are accurately classified. Combining Figs. 7, 9, and 10, it can be found that the main wrong predictions focus on the changes related to "vegetation_region" and "industrial_region," which have the largest number of samples among all categories in Hanyang dataset. In the first temporal image, it is evident that the main incorrect predictions involve the recognition of many "vegetation_region" scenes as "idle_region." The reason for this may lie in the fact that winter farmland lacks crops, and the main land covers in the field are bare soil, thus making "vegetation_region" highly similar to "idle_region." On the other hand, in the second temporal image, the main incorrect predictions involve the recognition of many "industrial_region" scenes as "idle_region" and "residential_region." The reasons for these phenomena may be due to some "industrial_region" scenes containing numerous black buildings and scattered goods, which closely resemble those in "idle_region." In addition, some "industrial_region" scenes consist of buildings and vegetation, with spatial relationships that are highly similar to those found in "residential_region" scenes.

C. Experiment 2: The WHMAVS Dataset

The WHMAVS dataset [7] was first introduced in [5] and then revised in [7]. The dataset covers the whole of the city of Wuhan, China, with bitemporal remote sensing images of a spatial resolution of 1.2 m, acquired in 2014 and 2016 from Google Earth. The bitemporal images are of size 47 537 × 38 100, which were cropped to 23 567 bitemporal image pairs of size 200 × 200. These images are categorized into 14 land-use/land-cover categories, which are 1—administration, 2—commercial_region, 3—water, 4—agricultural_region, 5—green_space, 6—transportation, 7—industrial_region, 8—residential_region 1, 9—residential_region 2, 10—residential_region 3, 11—road, 12—parking_lot, 13—bare land, and 14—playground. This means a potential of 196 scene change categories. The distribution of the number of bitemporal scene categories is presented in Fig. 11. The WHMAVS dataset presents a challenge due to its imbalanced scene distribution and a multitude of change types, as illustrated in Fig. 11. According to a ratio of 7:1:2, the cropped bitemporal pairs were divided into the training, validation, and test sets. For this dataset, 100 epochs mean 50 000 iterations.

TABLE III
QUANTITATIVE RESULTS OBTAINED ON THE WHMAVS DATASET. THE BEST AND SECOND-BEST ACCURACIES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Backbone	Method	OA_t1	OA_t2	OA_bi	OA_tr	KC
VGGNet-16	Baseline	90.79	92.49	91.01	87.42	85.49
	CorrFusionNet	92.38	93.38	92.51	89.31	87.65
	AGOS	92.02	92.94	91.81	88.74	86.98
	D-CNN	90.81	92.41	90.81	87.36	85.40
	Sphere loss	92.72	93.55	92.34	89.54	87.92
	SCC (proposed)	**95.48**	**95.27**	**95.46**	**93.17**	**92.08**
ResNet-50	Baseline	88.91	90.45	87.44	84.03	81.63
	CorrFusionNet	89.63	90.41	88.48	84.83	82.52
	AGOS	88.95	90.43	89.39	84.90	82.59
	D-CNN	90.69	92.15	89.58	86.68	84.64
	Sphere loss	90.43	91.51	89.75	86.30	84.21
	SCC (proposed)	**94.63**	**95.21**	**94.91**	**92.51**	**91.33**
DenseNet-121	Baseline	89.37	91.90	88.76	85.43	83.24
	CorrFusionNet	91.37	91.83	90.69	87.34	85.39
	AGOS	90.64	91.56	89.94	86.59	84.54
	D-CNN	89.75	90.67	90.26	85.89	83.72
	Sphere loss	91.68	93.45	91.51	88.61	86.86
	SCC (proposed)	**95.21**	**94.65**	**95.21**	**92.66**	**91.50**

"SCC" means the proposed SCC framework.

1) Quantitative Experimental Results: As shown in Table III, the proposed SCC framework with the three backbones outperforms the other methods in all the evaluation metrics. All the methods obtain their highest performance with VGGNet16. With VGGNet16, the proposed SCC framework outperforms the other methods in all the evaluation metrics, by 1.72%–2.85% in OA_t1 and OA_t2, 2.94%–4.64% in OA_bi, 3.62%–5.81% in OA_tr, and 4.11%–6.67% in KC. This is due to the suitable hyperparameter setting for VGGNet16 [5]. However, the proposed SCC framework still shows its superior SCD ability when using the commonly used DenseNet-121 and ResNet-50 backbones. For example, the proposed SCC framework with DenseNet-121 backbone outperforms the other methods by 1.21%–5.83% in OA_t1 and OA_t2, 3.70%–6.45% in OA_bi, 4.04%–7.22% in OA_tr, and 4.64%–8.26% in KC. Table IV shows the superiority of the proposed SCC framework from the perspective of binary SCD. Table IV also demonstrates that AGOS outperforms all other methods in recognizing changed scene pairs, but performs worse than the advanced methods in recognizing unchanged scene pairs.

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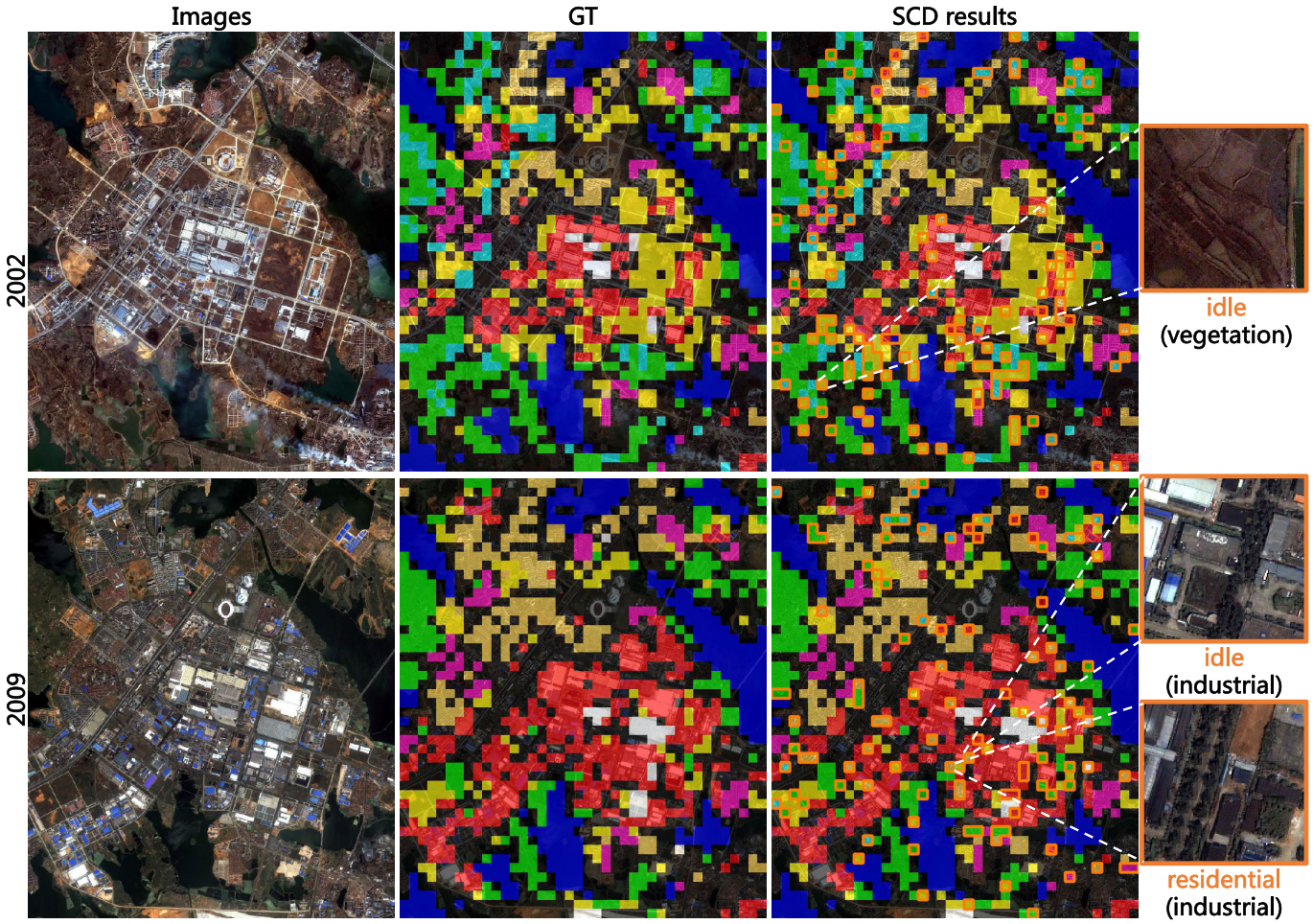


Fig. 10. Visualization of the SCD results of the proposed SCC framework with the DenseNet-121 backbone on the whole test region of the Hanyang dataset. Within the third column, the image patches enclosed by the orange box illustrate the incorrectly predicted scenes (■ undefined, □ parking, ■ water, ■ sparse_houses, ■ dense_houses, ■ residential_region, ■ idle_region, ■ vegetation_region, and ■ industrial_region). The region is of size 7200 × 6000, which covers 43.2 km^2.

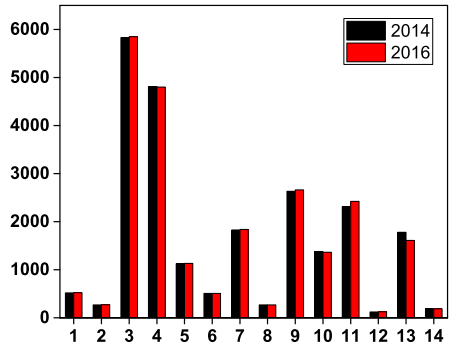


Fig. 11. Distribution of the number of bitemporal scene categories in the WHMAVS dataset.

TABLE IV
BINARY SCD RESULTS OBTAINED USING DENSENET-121 OBTAINED ON THE WHMAVS DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

	TP	FN	FP	TN
Baseline	45	15	515	4139
CorrFusionNet	45	15	424	4230
AGOS	**50**	**10**	464	4190
D-CNN	49	11	448	4206
Sphere loss	49	11	389	4265
SCC (proposed)	49	11	**215**	**4439**

Fig. 12 shows the confusion matrices for the scene classification results obtained by the two SCD methods with DenseNet-121 backbone, which shows that the accuracies of most of the scene classes obtained by the proposed SCC framework are higher than those of CorrFusionNet. Meanwhile, the error rates of the predictions of the proposed SCC framework are more balanced than those of CorrFusionNet. These experimental results are similar to the experimental results obtained on the Hanyang dataset. Among them, the accuracy rate of "administration," "transportation," "industrial_region," and "playground" increased by more than 10%. Notably, corresponding to the example of Fig. 1, the accuracy of the classification for "road" has significantly improved from 88.68% to 93.97%. In particular, compared to CorrFusionNet, which misclassifies 2.54% of "road" samples as "bare land," the proposed SCC framework achieves a much lower misclassification rate of only 0.63%. These experimental results consistently confirm the effectiveness of the proposed method in obtaining a strong and stable SCD ability for the Siamese network on the large-scale WHMAVS SCD dataset.

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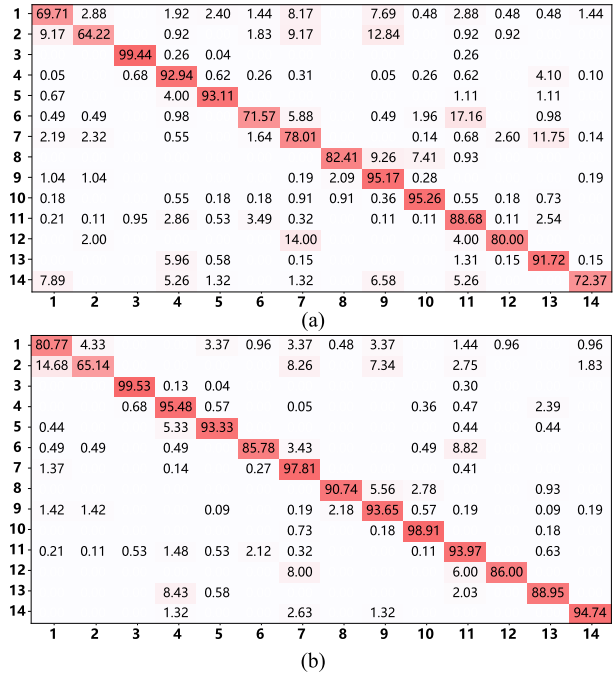


Fig. 12. Confusion matrices for the bitemporal scene classification results obtained on the test set of the WHMAVS dataset using (a) CorrFusionNet [5] and (b) proposed SCC framework, with DenseNet-121 backbone. 1—Administration, 2—commercial_region, 3—water, 4—agricultural_region, 5—green_space, 6—transportation, 7—industrial_region, 8—residential_region 1, 9—residential_region 2, 10—residential_region 3, 11—road, 12—parking_lot, 13—bare land, and 14—playground.

It can also be found in Table III that D-CNN with VGGNet16 performs worse than the baseline VGGNet16, while sphere loss with each backbone can obtain promising results on the large-scale WHMAVS SCD dataset. These experimental results are similar to those obtained on the Hanyang dataset.

2) Feature Analysis: In this section, visualization and statistical measures analysis were conducted on the learned features to understand them better. Here, the feature visualization and analysis of the AGOS method are not displayed because AGOS fuses the logits of features from different branches and does not provide a final feature representation for a single scene image.

The scene features were visualized using t-SNE [56] and the results are shown in Fig. 13. It can be observed that the feature clusters of "industrial_region," "residential_ region 2," and "road" are more compact for CorrFusionNet compared to the baseline methods. This is likely due to the presence of a large number of unchanged instances of these scene categories in the dataset, enabling CorrFusionNet to effectively learn the intraclass relationships among these scenes across different time phases. On the other hand, D-CNN and Sphere loss, by directly modeling the feature relationships, have a better separation between features of different scene categories overall compared to the baseline methods and CorrFusionNet. Compared to these methods, the SCC method learns more compact feature clusters for each scene category, with lower confusion levels among different categories. In particular, most of the playground category features can significantly separate from the feature clusters of other scene categories. This observation is consistent with the quantitative indicators for the playground category in the confusion matrix shown in Fig. 12.

The learned change representations and scene features were further evaluated by analyzing their statistical measures, i.e., the average intraclass distance D^{intra}, the average interclass distance D^{inter}, and the interclass distance standard deviation (std) S^{inter}, which measures the geometry configuration of the feature distribution. A smaller value of S^{inter} indicates better interclass uniformity and a more balanced feature space, which in turn corresponds to an improvement in generalization performance [11], [57], [58]. The three measures are defined as follows:

$$D^{\text{intra}} = \frac{1}{X}\sum_{x=1}^{X} d_x^{\text{intra}},\ D^{\text{inter}} = \frac{1}{X}\sum_{x=1}^{X} d_x^{\text{inter}} \tag{18}$$

$$S^{\text{inter}} = \sqrt{\frac{1}{X-1}\sum_{x=1}^{X}\left(D^{\text{inter}} - d_x^{\text{inter}}\right)^2}. \tag{19}$$

Here, X represents the number of classes present in either the scene or change types. Specifically, when computing the measures for scene features, X is equivalent to the number of scene classes, denoted as N. Conversely, when computing the measures for change representations, X equals to the number of change types, represented as N^2. The intraclass distance d_x^{intra} of the xth scene class or change type was calculated by taking the mean distance between the intraclass features and the feature center $\mathbf{c}_x$ of the xth class, while the interclass distance d_x^{inter} for the xth class is defined as the distance between the centers of different classes

$$d_x^{\text{intra}} = \frac{1}{u_x}\sum_{i=1}^{u_x} ||\mathbf{r}_i - \mathbf{c}_x||_2^2 \tag{20}$$

$$d_x^{\text{inter}} = \frac{1}{X-1}\sum_{x=1, x\neq y}^{X} ||\mathbf{c}_x - \mathbf{c}_y||_2^2 \tag{21}$$

$$\mathbf{c}_x = \frac{1}{u_x}\sum_{i=1}^{u_x} \mathbf{r}_i \tag{22}$$

where u_x is the number of samples of the xth class, $\mathbf{r}_i$ is the ith scene feature or change representation of the xth class in the test set. Here, the change representation is defined as the concatenation of the bitemporal features, which corresponds to the proposed method. The experimental results are shown in Table V.

Table V rows 2–6 show that these methods significantly increase the average interclass distance D^{inter} of the change representations. While the proposed SCC framework does not achieve the minimum average intraclass distance D^{intra} or maximum average interclass distance D^{inter}, it does attain the minimum interclass distance std S^{inter}. The experimental results demonstrate that the proposed SCC framework improves the interclass distance and reduces its standard deviation, resulting in improved SCD performance. Meanwhile, the experimental results in Table V rows 7–11 lead to a similar conclusion regarding the scene features.

The reason why the proposed SCC framework can effectively reduce confusion among bitemporal feature clusters lies

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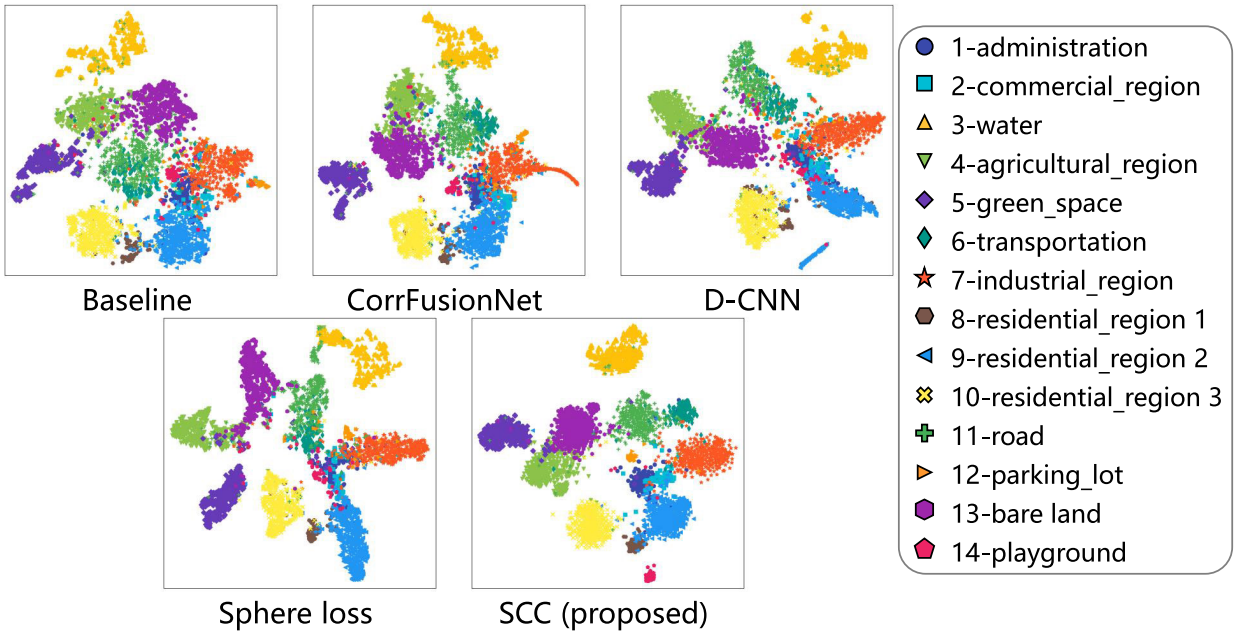


Fig. 13. t-SNE visualization of the scene features using DenseNet backbone on the WHMAVS dataset.

TABLE V
STATISTICAL MEASURES OF DIFFERENT METHODS USING DENSENET-121 OBTAINED ON THE WHMAVS DATASET

	Method	D^{intra}	D^{inter}	S^{inter}
change representations	Baseline	13.63	29.08	11.38
	CorrFusionNet	126.76	**311.39**	1038.90
	DCNN	15.84	23.68	40.62
	Sphere loss	**12.11**	41.49	11.18
	SCC (proposed)	19.87	43.92	**9.53**
scene features	Baseline	13.78	20.48	10.29
	CorrFusionNet	36.53	**46.53**	66.13
	DCNN	20.65	25.96	40.73
	Sphere loss	**11.41**	29.29	9.56
	SCC (proposed)	20.36	31.64	**6.55**

in considering the contrastive representation of changes from the perspective of bitemporal scenes as a whole. By imposing stronger temporal constraints on scene features simultaneously, the SCC method enhances the overall ability to describe changes in bitemporal scene features, thereby effectively guiding the improvement in discriminability among scene features.

3) Class Activation Mapping (CAM): To understand how the SCC learning uses bitemporal scene features to describe changes, Grad-CAM [59] was used to visualize the CAM of the features output by the Siamese networks, to reveal the discriminative regions of the scene features. The CAM experimental results were obtained by the five methods, each with a DenseNet-121 backbone, and are exhibited in Fig. 14, where the lighter colors denote stronger CAM responses.

The first and second rows and the third and fourth rows of the figure show examples of two scenes that do not change and change, respectively. From the first and second rows of Fig. 14, it can be seen that larger imaging differences cause the existing methods to fail to accurately locate the discriminative region in both unchanged scenes, resulting in false alarms for scene changes. In the third row of Fig. 14, the scene category of the first temporal image is "bare land," but it is incorrectly classified as a "parking lot" by most existing methods. The reason for this is that the excavated land in bare land presents a grid-like structure, which is very similar to the densely arranged cars in a parking lot. In the fourth row of Fig. 14, the scene category of the second temporal image is "road," but a large amount of bare soil causes great interference to existing methods, making them incorrectly classify it as "bare land." However, by observing the CAM results of the proposed SCC method, it can be found that the features learned by the SCC method can accurately locate the same discriminative region for unchanged scenes, and can locate different discriminative regions for changed scenes before and after the change, thus correctly identifying the above challenging types of scene changes.

V. ABLATION STUDY

In this section, we describe the ablation study conducted with the different parts of the proposed method and their hyperparameters, to evaluate their influence on the SCD performance. The experiments were conducted using the DenseNet-121 backbone and the binary SCD metric OA_bi and the from-to SCD metric OA_tr on the Hanyang and WHMAVS datasets. The experimental settings of the ablation study were the same as Section IV-A1. The batch size was set to $N = 32$, which includes 32 bitemporal scene pairs.

A. Effect of the Different Parts in the SCC Framework

The proposed SCC learning method includes two basic designs, i.e., the MVC projector, which provides the local and global-view change projections, to exploit the change information for the joint bitemporal features, and the PeCA strategy, which increases the quantities and classes of the change projection for the SCC learning. Table VI row 2 shows the experimental results predicted by the independent PCC

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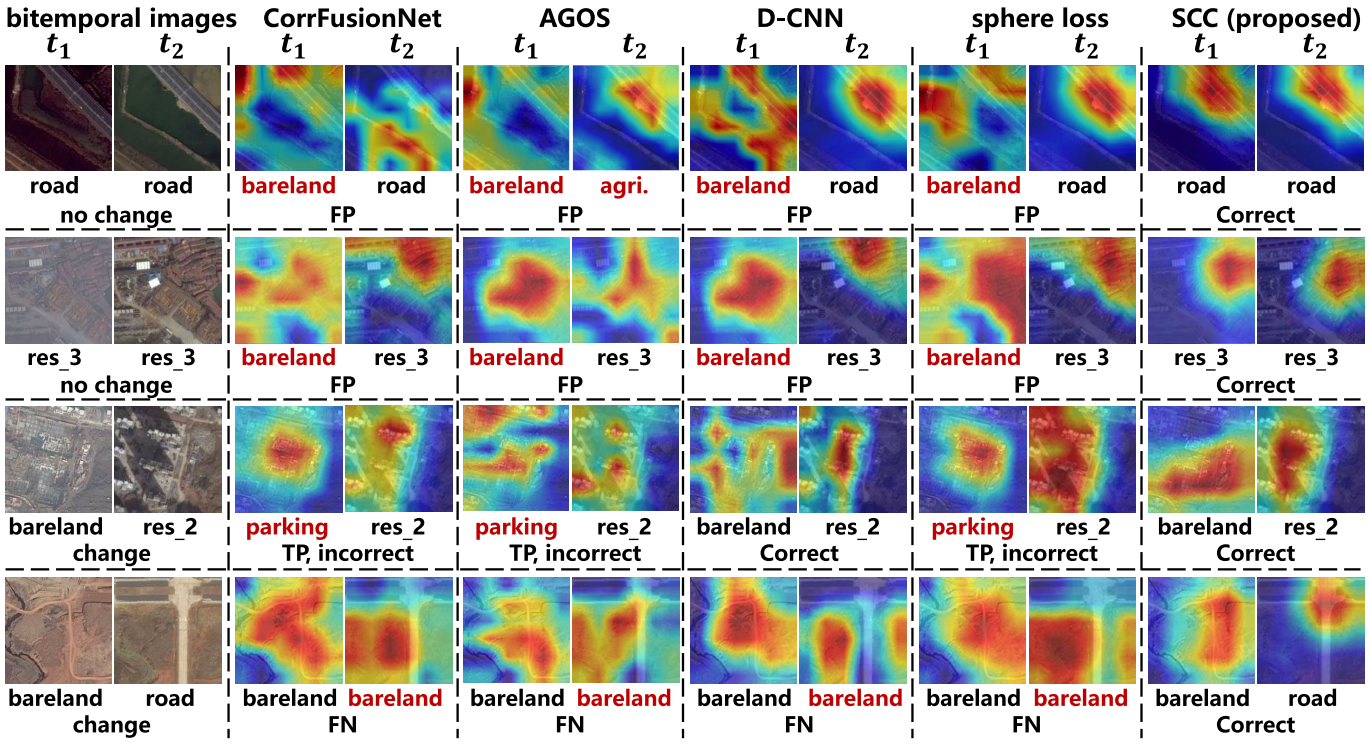


Fig. 14. Visualization of the CAM using Grad-CAM with DenseNet-121 backbone on five scene pairs. "FP" and "FN" denote the false positives and false negatives of change, respectively. "Correct" means that the semantic change type is correctly predicted. The incorrect single-temporal scene predictions are highlighted in red.

TABLE VI

EFFECT OF DIFFERENT PARTS OF THE SCC FRAMEWORK WITH DENSENET-121 ON THE HANYANG AND WHMAVS DATASETS. THE BEST ACCURACIES ARE BOLD-FACED

Method		Hanyang dataset		WHMAVS dataset	
		OA_bi	OA_tr	OA_bi	OA_tr
DenseNet121		86.00	76.57	88.76	85.43
+ MVC projector	global-view only	87.14	78.48	94.46	92.15
	local-view only	86.86	77.52	94.25	92.11
	global + local (multi-view)	87.24	79.33	94.55	92.19
+ MVC projector + PeCA strategy (+ SCC)		**88.48**	**80.48**	**95.21**	**92.66**

baseline with DenseNet-121, while Table VI rows 3 and 4 show the experimental results obtained by the SCC framework without pseudo change projections and the whole SCC framework, respectively.

1) MVC Projector: Compared with the DenseNet-121 baseline (Table VI row 2), the SCC learning with only real change projections (Table VI row 5) improves the OA_bi by 1.24% and the OA_tr by 2.76% on the Hanyang dataset, and improves the OA_bi by 5.79% and the OA_tr by 6.76% on the WHMAVS dataset. These experimental results verify the effectiveness of both the MVC projector and the SCC learning, which can benefit more from a large-scale dataset than from a small-scale dataset. Table VI rows 3–5 show that the global-view and local-view change projections play different roles in SCC learning. For the small-scale Hanyang dataset, incorporating both global-view and local-view change projections leads to significant enhancements in OA_tr by 0.85% and 1.81%, respectively, as compared to utilizing solely global-view or local-view change projections.

2) PeCA Strategy: Comparing Table VI rows 3 and 4, it can be found that the PeCA strategy enhances the SCC learning using only real change projections by 1.24% in OA_bi and 1.15% in OA_tr on the Hanyang dataset, and 0.66% in OA_bi and 0.47% in OA_tr on the WHMAVS dataset. These experimental results indicate that the PeCA strategy is more effective on a small dataset than on a large dataset. The reason for this may be that the pseudo bitemporal pairs are not independent identically distributed (i. i. d.), like the color jitter strategy of image augmentation, when compared with the real pairs. Therefore, when a large-scale dataset includes enough SCD information, the SCC learning can only benefit slightly from the PeCA strategy.

B. Different Operators to Obtain Change Projections

In Section III-B, global-view and local-view change projections are obtained from bitemporal feature vectors and bitemporal local vectors through concatenation operation, as in (9) and (11), respectively. In this section, different potential operations, i.e., element-wise addition, subtraction, multiplication, and weighted sum, to obtain the change projections were explored and compared. In this context, weighted sum refers to the process of adding together bitemporal vectors, with each vector multiplied by a corresponding learnable scalar parameter.

Table VII demonstrates that the concatenation operator does not achieve the highest OA_bi and OA_tr scores compared to the other operators. However, when examining the abilities

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TABLE VII
SCD PERFORMANCE OF DENSENET-121 WITH THE SCC FRAMEWORK USING DIFFERENT OPERATORS TO OBTAIN CHANGE PROJECTIONS. THE BEST VALUES ARE BOLD-FACED

Dataset	Operator	OA_bi	OA_tr	TP	FN	FP	TN	OA_ch	OA_no
Hanyang dataset	addition	87.90	78.57	383	30	97	540	92.74	84.77
	subtraction	88.86	80.29	376	37	**80**	**557**	91.04	**87.44**
	multiplication	**89.14**	**80.95**	383	30	84	553	92.74	86.81
	weighted sum	88.19	78.95	383	30	94	543	92.74	85.24
	concatenation	88.48	80.48	**386**	**27**	94	543	**93.46**	85.24
WHMAVS dataset	addition	94.08	91.79	48	12	267	4387	80.00	94.26
	subtraction	95.16	92.83	**49**	**11**	217	4437	**81.67**	95.34
	multiplication	94.74	92.68	45	15	236	4418	75.00	94.93
	weighted sum	95.18	**92.98**	47	13	**214**	**4440**	78.33	**95.40**
	concatenation	**95.21**	92.66	**49**	**11**	215	4439	**81.67**	95.38

to detect changes and no changes, the metrics OA_ch = TP/(TP + FN) and OA_no = TN/(TN + FP) are introduced, respectively. In this context, it can be observed that using concatenation has the best effect on identifying changes, but the ability to identify no changes is inferior to other operators. The key reason for this is that each element of the change projections obtained by the other operators is calculated jointly with the elements of the original bitemporal vectors. As a result, individual elements within the change projection differ from those in the bitemporal vectors, leading to a loss of information pertaining to single temporal scenes while emphasizing the overall information present within bitemporal scenes. Therefore, under the data distribution where the number of nonchange samples is much larger than that of change samples, the results obtained by these operators are better than the DenseNet baseline, and their recognition rate for unchanged scenes is high. In contrast, the concatenation operator not only enables the change projection to describe the ability of scene pairs but also does not destroy the discriminative information of each scene, making it easier to distinguish between changes and no changes in the change projection space.

Considering the importance of identifying changes in SCD, the concatenation operator was adopted for all experiments.

C. Effect of the Pseudo Change Projections

The experimental results in Table VIII show the effect of different amounts of pseudo change projections. The experimental results presented in Table VIII row 3 indicate that the SCC learning was performed only with real change projections. The experimental results presented in the last row of Table VIII indicate that the crossing the geographical locations was performed on all the bitemporal pairs, and correspond to the experimental results obtained by the proposed SCC framework in Tables I, III, and VI. In each experiment of Table VIII rows 4–10, the different numbers of real bitemporal scene pairs were crossed to obtain pseudo change projections. For example, in Table VIII row 5, 2×32 real change projections were obtained from 32 real bitemporal scene pairs, while $2 \times (8^2 - 8)$ pseudo change projections were obtained from eight randomly selected real bitemporal scene pairs. From

TABLE VIII
SCD PERFORMANCE OF DENSENET-121 WITH THE SCC FRAMEWORK USING DIFFERENT AMOUNTS OF PSEUDO CHANGE PROJECTIONS. THE BEST AND SECOND-BEST VALUES ARE BOLD-FACED AND UNDERLINED, RESPECTIVELY

The number of change projections		Hanyang dataset		WHMAVS dataset	
Total	2×(real+pseudo)	OA_bi	OA_tr	OA_bi	OA_tr
2×32	$2 \times (32 + 0)$	87.24	79.33	94.55	92.19
2×44	$2 \times (32 + 4^2 - 4)$	87.33	78.10	94.87	92.47
2×88	$2 \times (32 + 8^2 - 8)$	87.71	77.43	**95.42**	**93.15**
2×164	$2 \times (32 + 12^2 - 12)$	86.29	78.10	95.18	<u>92.87</u>
2×272	$2 \times (32 + 16^2 - 16)$	86.57	77.52	94.85	92.51
2×412	$2 \times (32 + 20^2 - 20)$	88.10	<u>80.19</u>	94.87	92.60
2×584	$2 \times (32 + 24^2 - 24)$	**88.57**	79.71	94.80	92.45
2×788	$2 \times (32 + 28^2 - 28)$	88.38	79.71	95.08	92.70
2×1024	$2 \times (32 + 32^2 - 32)$	<u>88.48</u>	**80.48**	<u>95.21</u>	92.66

Table VIII rows 4–11, it can be observed that the pseudo change projections obtained by the PeCA strategy can enhance the SCC learning, but the highest accuracy values do not always appear with the largest amounts of pseudo change projections, which demonstrates the different significance of the obtained pseudo change projections.

D. Change Projection Dimension

Fig. 15 illustrates the SCD performances, namely, OA_bi and OA_tr, at different change projection dimensions. As shown in Fig. 15(a), on the Hanyang dataset, the SCD performances achieve their highest values when the change projection dimension is 512, and its performance is significantly higher (+3%) than that obtained by other dimension values. At the same time, the SCD performances obtained by other dimension values are relatively close. As shown in Fig. 15(b), on the WHMAVS dataset, although the best OA_bi and OA_tr are achieved at a change projection dimension of 256 and 512, respectively, the SCD performances do not vary significantly with the changes projection dimension. The impact of the change projection dimension on SCD performances in the WHMAVS dataset is similar to that of the

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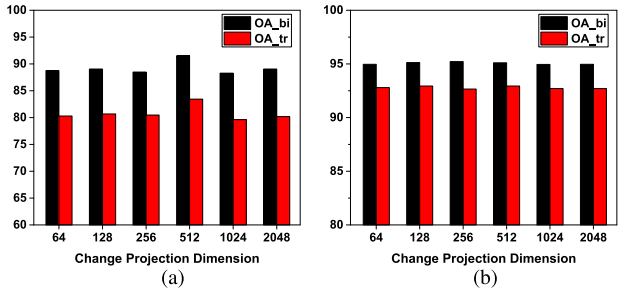


Fig. 15. SCD performances with different change projection dimensions. (a) Hanyang dataset. (b) WHMAVS dataset.

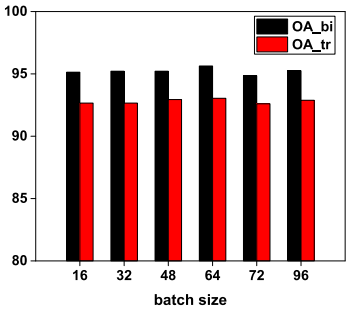


Fig. 16. SCD performances with different batch sizes on the WHMAVS dataset.

projection dimension of a single image on the classification performance on the ImageNet dataset [30].

The reason for the differences observed in the ablation experiment of the change projection dimension on the Hanyang, WHMAVS, and ImageNet datasets may be due to the difference in dataset size. Generally speaking, the smaller the projection dimension, the more the main discriminative dimensions of the features are emphasized, and the more detailed information is ignored. In this context, supervised contrastive learning benefits from large-scale training datasets like WHMAVS and ImageNet, which enable it to capture both main discriminative information. Therefore, the specific number of predefined projection dimensions becomes less important. However, for small-scale training datasets such as the Hanyang dataset, learning high-quality features automatically is more challenging. In such cases, the predefined projection dimension plays a critical role.

In this study, to verify the proposed SCC learning method, the projection dimension of a single image is set to the typical value of 128 [25], [30], which means that the change projection dimension is set to 256, and promising SCD performances compared to baselines and other advanced comparison methods can be achieved.

E. Batch Size

Due to the limited size of the training set in the Hanyang dataset, batch size ablation experiments were conducted on the WHMAVS dataset. As shown in Fig. 16, batch size was varied from 16 to 96, which means 32–192 scene images. Batch sizes larger than 96 were not validated due to the limitations in GPU memory. From the experimental results in Fig. 16, it can be seen that the SCD performances are not significantly affected by the batch size. This conclusion is inconsistent with the finding that supervised contrastive learning shows

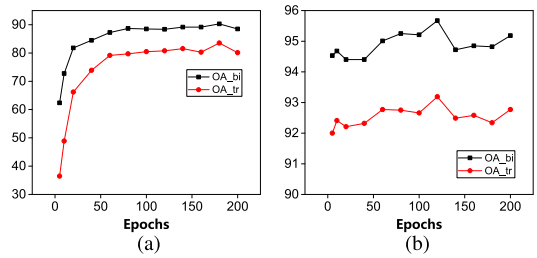


Fig. 17. SCD performances with different epochs. (a) Hanyang dataset. (b) WHMAVS dataset.

improved performance with increased batch size on the ImageNet dataset [25]. The reason for this discrepancy may lie in the differences in the number of classes between the ImageNet and WHMAVS datasets. When the batch size is 32, a single batch of images can not cover the 1000 categories in the ImageNet dataset, but it may fully cover the 14 scene categories and partially cover the 196 scene change types in the WHMAVS dataset. Therefore, the proposed SCC learning is not very sensitive to batch size. However, if there are more and finer scene categories, the proposed SCC learning may become more sensitive to the batch size.

F. Epochs

The experimental results of the proposed SCC method on the two datasets under different epochs are depicted in Fig. 17. It can be observed that when the number of epochs is less than 100, the values of OA_bi and OA_tr metrics achieved on both datasets exhibit a noticeable increase as the number of epochs increases. However, when the number of epochs exceeds 100, the values of OA_bi and OA_tr metrics achieved on the Hanyang dataset show a slight increase with the number of epochs increasing. On the other hand, there is no clear correlation between the number of epochs and the SCD performances achieved on the WHMAVS dataset. The disparity can be attributed to the variation in training data volume between the two datasets. The Hanyang dataset has only 190 pairs of bitemporal images, while the WHMAVS dataset has 16 490 pairs. Consequently, the SCD performances obtained on the Hanyang dataset are more sensitive to the number of epochs compared to those on the WHMAVS dataset.

G. Hyperparameters

λ, which is the predefined hyperparameter to balance the SCC learning and the bitemporal scene classification in (7) of the joint loss, was set to different values with DenseNet-121 to verify the effect of the SCC learning. As shown in Fig. 18(a), when λ is set to 0.1–0.9, the SCC framework with DenseNet-121 backbone can achieve promising OA_bi and OA_tr scores on the Hanyang and WHMAVS datasets. Considering that OA_bi and OA_tr gradually degrade as the λ value becomes larger than 0.5, we fixed λ at 0.5 in all the experiments. When λ is set to 0, the SCC framework with DenseNet-121 backbone is only trained to learn bitemporal scene classification, substantially reducing the SCD performance.

q is the predefined hyperparameter to control the local information extraction on a feature map of size $H \times W \times C$

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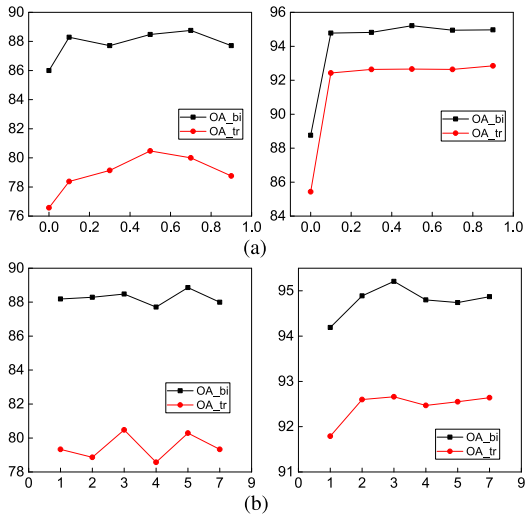


Fig. 18. Analysis of (a) λ to balance the SCC learning and the bitemporal scene classification in (7) of the joint loss with DenseNet-121 and (b) hyper-parameter q of the GeM pooling, as in (10) in the MVC projector. (Left) Hanyang dataset. (Right) WHMAVS dataset.

in (10) of the GeM pooling in the MVC projector. Here, $q = 1$ means that the GeM pooling degenerates to the global average pooling. As shown in Fig. 18(b), with the increase of q, the SCD performance increases at first but then quickly falls. When $q = 3$, which happens to be the classical value in [52], nearly all SCD performances reach their maximum values. This improves the proposed method with $q = 1$ by 0.29% in OA_bi and 1.15% in OA_tr on the Hanyang dataset, as well as by 1.02% in OA_bi and 0.87% in OA_tr on the WHMAVS dataset. When q is greater than 7, the model output became "Not a Number" (NaN) for each dimension of the logits. This is due to the fact that a larger value of q makes the GeM pooling function approach the Max pooling function. As a result, gathering together local-view change projections and global-view change projections of the same change type in the SCC learning is almost equivalent to bringing the outputs of max pooling and average pooling closer, leading to the model collapse.

VI. Conclusion

In this article, the SCC framework has been proposed to handle the issue of insufficient change representation discrimination. The SCC framework includes bitemporal scene classification with a regularization term of SCC learning based on the contrastive learning technique. The SCC learning asks a Siamese network to contrast the change projections, to guide the joint bitemporal features to be discriminative change representations. The quantitative experimental results obtained on two public datasets consistently demonstrated the effectiveness of the proposed SCC framework. The class activation map visualization showed that the bitemporal features learned by the SCC learning can highlight the same regions for the unchanged bitemporal scene pairs, and highlight accurate discriminative regions for the changed bitemporal scene pairs, demonstrating the ability of the joint bitemporal features to describe various changes. In our future work, considering that different classes and amounts of pseudo-change projection have different effects on SCD, we will work on designing a more dedicated strategy to construct and select pseudo-change projections for SCC learning. Furthermore, the SCD task is confronted with the challenge of a pronounced imbalance in sample distribution, which can exacerbate the difficulty of contrastive learning [60]. Consequently, investigating contrastive learning approaches that take into account the imbalanced distribution of scene changes represents a promising avenue for enhancement.

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